Myungsin Cho

Teaching Statement

I have taught mathematics at different levels and in a range of formats. My experience spans general-education courses for nonmajors, service courses such as calculus, proof-based linear algebra, and applied mathematics for STEM students. I have taught in traditional classrooms and large lectures with more than 100 students, as well as in flipped classrooms and through synchronous online lectures.

Across these settings, I want students to experience mathematics as one connected story. I build my courses around concrete questions, helping students see why each new concept is needed and how ideas fit together. My goal is for them to build fundamental skills and use that understanding to explain their reasoning, evaluate arguments, and explore questions of their own.

Connecting Questions and Concepts

I adapt my teaching to students' preparation and to the purpose of each course. As a co-instructor for Math of Beauty and Decision, I taught nonmajors through everyday examples involving voting, symmetry, and games. My aim was to help students discover mathematics in familiar situations and see how mathematical theories describe them. In core and applied courses, I emphasize the foundations students will need later, while encouraging those with stronger preparation to explore the ideas in greater depth.

In Analysis and Optimization, students come from various STEM fields, and many have never written a formal proof. I redesigned some homework assignments as guided worksheets that help students understand the structure of a proof while asking them to complete the reasoning themselves. Students have described proofs as more accessible and intuitive with this support. I provide a starting point from which they can develop the ability to construct arguments independently.

During class, I invite questions and ask students to anticipate the next step or explain why an argument should work. Their responses and written work help me decide when to review an idea, provide another example, or ask them to work more independently. I also learn students' names and majors, encourage them to attend office hours, and offer academic guidance when needed. These interactions help create an inclusive environment in which students feel comfortable asking questions and engaging with mathematics.

I organize homework, quizzes, and exams as connected parts of an ongoing learning process. This gives students repeated opportunities to work with core ideas and use their earlier work and assessment results to prepare for later assessments. I also provide opportunities to revisit mistakes and recover points, encouraging students to continue working after they receive a grade. I continually reconsider how these assessments fit together to support regular study and give students opportunities to demonstrate improvement.

Supporting Students' Learning

I adapt my teaching to students' preparation and to the purpose of each course. As a co-instructor for Math of Beauty and Decision, I taught nonmajors through everyday examples involving voting, symmetry, and games. My aim was to help students discover mathematics in familiar situations and see how mathematical theories describe them. In core and applied courses, I place greater emphasis on the skills and concepts students will need in their later studies.

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I want assessment to support learning beyond the moment students receive a grade. I organize homework, quizzes, and exams as connected parts of an ongoing learning process. This gives students repeated opportunities to work with core ideas and use their earlier work and assessment results to prepare for later assessments. I also provide opportunities to revisit mistakes and recover points, encouraging students to continue working after they receive a grade. I continually reconsider how these assessments fit together to support regular study and give students opportunities to demonstrate improvement throughout the course.

Developing Independent Mathematical Inquiry

I ask students to explain what they understand and why their reasoning works. In Linear Algebra, I used an oral final examination that gave students practice explaining what they knew as they prepared for it. Their explanations also helped me assess their understanding more clearly. I see this ability to communicate mathematical reasoning as an important part of learning to work independently.

Students also need to evaluate arguments and answers for themselves. When I discuss AI with students, I emphasize that learning mathematics is essential to using these tools properly and responsibly. Mathematical understanding allows them to examine assumptions, identify gaps in reasoning, and verify conclusions. I want students to develop the judgment to decide whether an answer is justified, including answers produced by AI, and to take responsibility for the arguments they present.

I also teach students to recognize questions that lead to further exploration. Questions about existence, uniqueness, and classification arise naturally from the objects we study, and I emphasize them as ways to continue thinking beyond an assigned problem. I see undergraduate research as an opportunity for students to develop and investigate such questions themselves. I would help students choose questions about familiar mathematical objects and refine them into problems they can begin to explore. Examples and calculations could lead to conjectures, proofs, or counterexamples, with new concepts introduced as they become useful. My role would be to help students find manageable directions while giving them increasing responsibility for the investigation and for explaining their findings.

I will continue to refine my teaching in response to students' needs and experiences. Across undergraduate courses and individual mentoring, I aim to help students develop the understanding and confidence to think mathematically for themselves.